Adversarial Environment of Machine Learning

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Adapted from Daniel Lee

Brief Overview

- Adversarial Environment
- Data Poisoning
- Adversarial Examples

Adversarial Environment

Machine Learning

- Models based on data and statistics
- Assumptions
  - Data is neutral
  - Noise is stochastic
- Adversarial Environment
  - Data is crafted
  - Noise is crafted

Quick Example

Attacker

- Email “I am spam”
- Email text: “I am junk mail”
- Email text: “I am not ham”
- Email image:

- Email image:

Defender

- Ignore “spam”
- Ignore synonyms
- Ignore synonym phrases
- Detect images
- Detect better

Phases of Statistical Modeling

- Model building
- Training
  - Gathering data
  - Collecting data
  - Learning from data
- Testing
  - Held-out data
  - Live data
  - Representative

Attack Types

- Data Poisoning
  - Attack on trainingdata
  - Makes classifiersunreliable
  - Decrease Effectiveness
- Adversarial Examples
  - Attack on test data
  - Untargeted Attacks
  - Targeted Attacks

Data Poisoning

Attacking SpamBayes: Compromising a Statistical Spam Filter

- U.C. Berkeley work on Adversarial Environments
- Attack Open Source Spam Detection tool (SpamBayes)
- Access to Training Data
  - Assume user retrains
- Two kinds of attacks including Dictionary Attack

How it trains?

- Through supervised learning
- Learns "SPAM" words and "HAM" words
- Weights the value of different words

- "New **Drug** on **discount**!"
- $0.5 + 0.9 = 1.4$
- "Can you **send** the **spreadsheet**?"
- $0.1 + -0.7 = -0.6$

DRUG	DISCOUNT	SPREADSHEET	CANDLE	SEND
+0.5	+0.9	-0.7	-0.3	0.1

Attacks

- Dictionary attack
  - Send English dictionary
  - Retrained and makes benign words seem malicious
  - Makes classification unreliable
- Find rare words
  - Educated Guess of rare, but usually HAM words.
  - Causes SPAM to be missed

Adversarial Examples

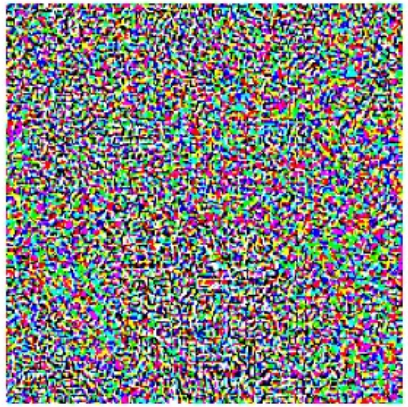
“Intriguing properties of neural networks”

- Vulnerability to perturbations.
  - Imperceptible to humans
  - Misclassification with higher confidence
- Transferability of models.
  - Don’t need original
  - Black-box attack

Szegedy, Christian, et al. "Intriguing properties of neural networks." *arXiv preprint arXiv:1312.6199* (2013).

Perturbation

57.7% panda



8.2% roundworms

99.3% gibbon

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and harnessing adversarial examples." *arXiv preprint arXiv:1412.6572* (2014).

Attacking NLP

Too Difficult?

- Relatively, images are simple
  - Natural language more complex.
- Research making progress.
  - For attacks and defenses
  - Adversarial Example Training

Neural Image Captioning

- Not 100% domain ofnatural language
- goal: Attack captioning systems with Adversarial Examples
- method: Show-and-Fool
  - devise a loss function to maximize output with minimal distortion
- Attack Types
  - Targeted Caption Method
  - Targeted Keyword Method
- result:
  - Targeted Caption: 95.8%
  - Target Keyword: 97.1%, 97.5%, 96.0% (1-, 2-, 3-keyword)

Chen, Hongge, et al. "Attacking Visual Language Grounding with Adversarial Examples: A Case Study on Neural Image Captioning." *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vol. 1. 2018.

Actual Natural Language

- Due to complexity of natural language new method.
- Image perturbs pixels not words
  - Inputs not continuous.
  - Perturbations more noticeable.
- Sample Results ***changed*** *[original]*

  This movie had ***horrific*** *[terrible] acting,* ***horrific*** *[terrible] plot, and* ***horrific*** *[terrible] choice of actors. (Leslie Nielsen …come on!!!) the one part I* ***regarded*** *[considered] slightly funny was the battling FBI/CIA agents, but because the audience was mainly* ***youngsters*** *[kids] they didn't understand that theme.*

- *Originally: Negative 78.0% confidence and now 59.8% Positive sentiment.*

Alzantot, Moustafa, et al. "Generating Natural Language Adversarial Examples." *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 2018.

Actual Natural Language (cont.)

- Unique Approach
- Perturb function.
  - Creates new sentence.
  - Similar meaning.
  - Closer to target.
- Genetic Algorithm
  - Several new sentences.
  - Selects new generation based on fitness of model output.
  - Employs crossover and mutation before next iteration.

Benefit of Adversarial Training

- Neural Machine Translation
  - Uses traditional GAN architecture.
  - Generator employs lexical level perturbations.
  - Discriminator must capture original intent.
- Joint Modelling (NER and relation extraction)
  - Goal: Find NER and mark relations between.
  - Words: concatenated character embedding and word embedding.
  - Perturbs concatenated embedding
  - Show that it improves Neural Network based models

Cheng, Yong, et al. "Towards Robust Neural Machine Translation." *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vol. 1. 2018.
Bekoulis, Giannis, et al. "Adversarial training for multi-context joint entity and relation extraction." *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 2018.

Questions

THANK YOU!